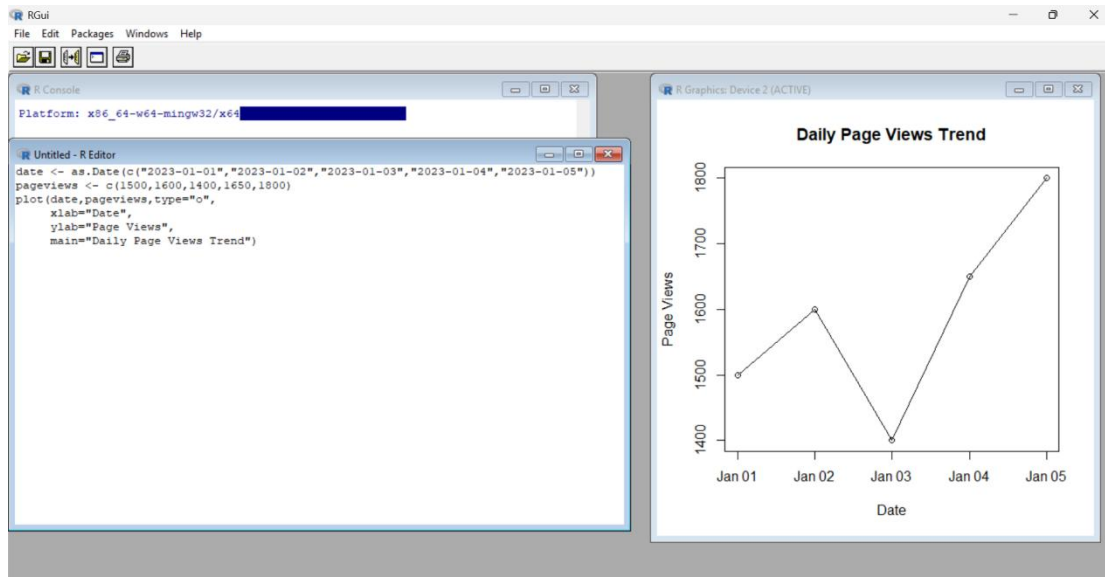
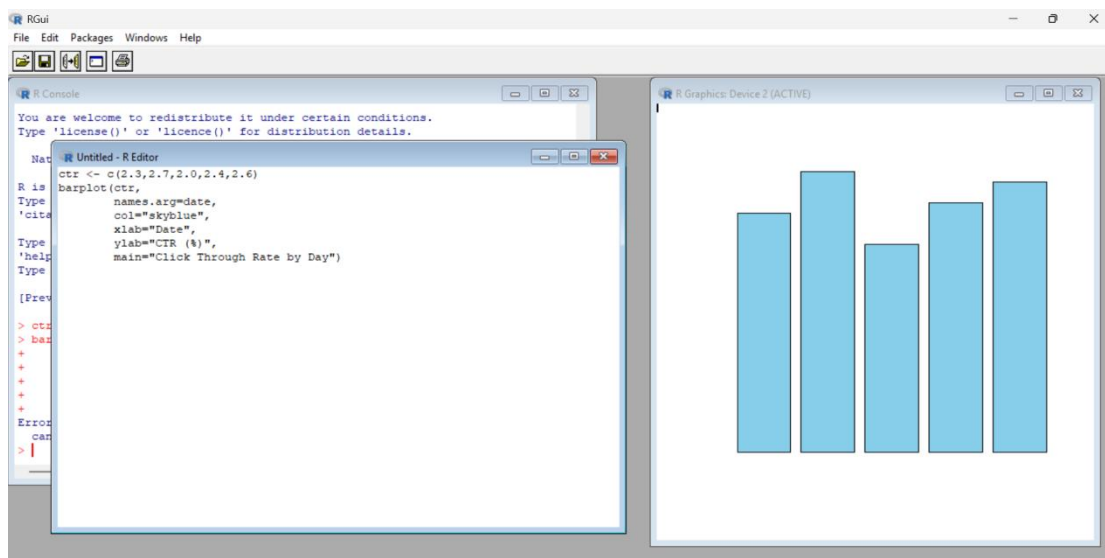


Set 5: Website Analytics

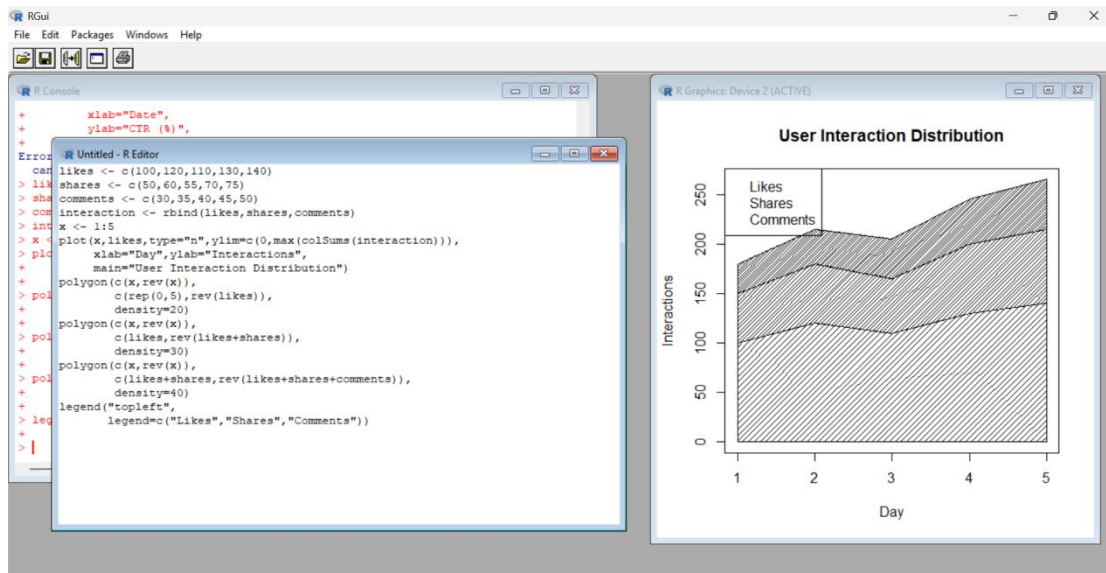
1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.



2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

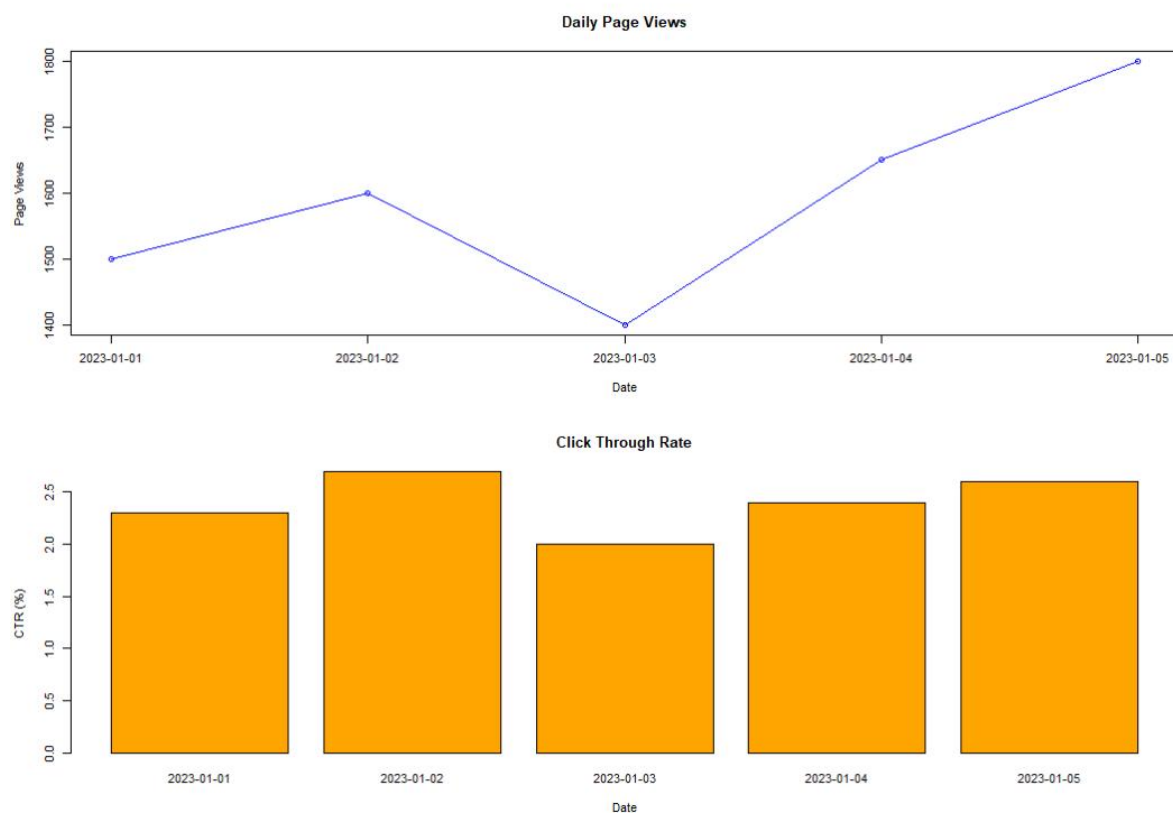


3. Develop a stacked area chart to display the distribution of user interactions (likes, shares, comments) on a website.



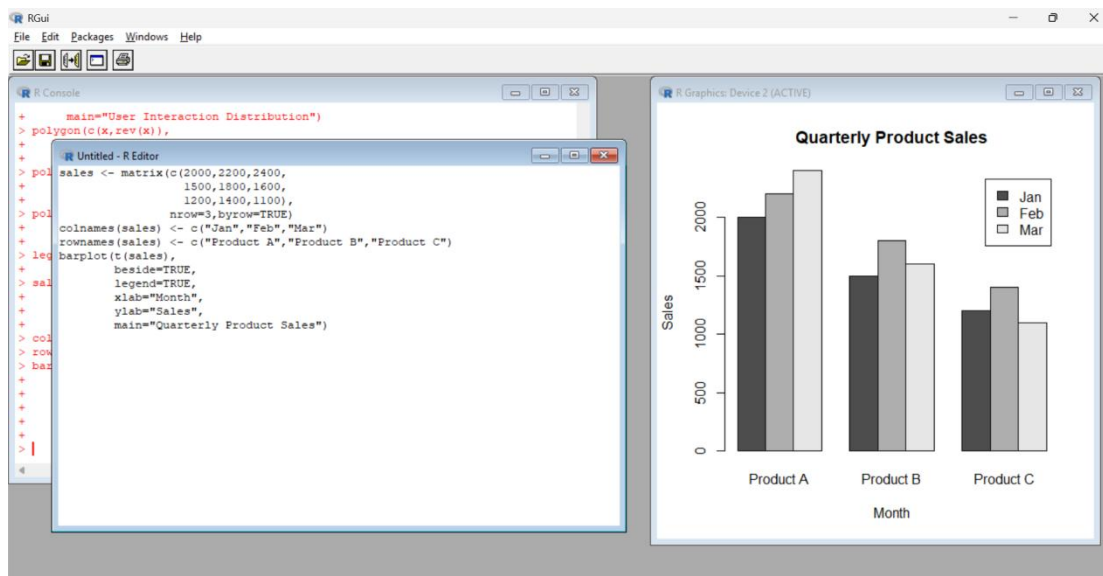
4. Build a Tableau dashboard with the line chart and bar chart for interactive exploration of website traffic data.

Website Traffic Dashboard

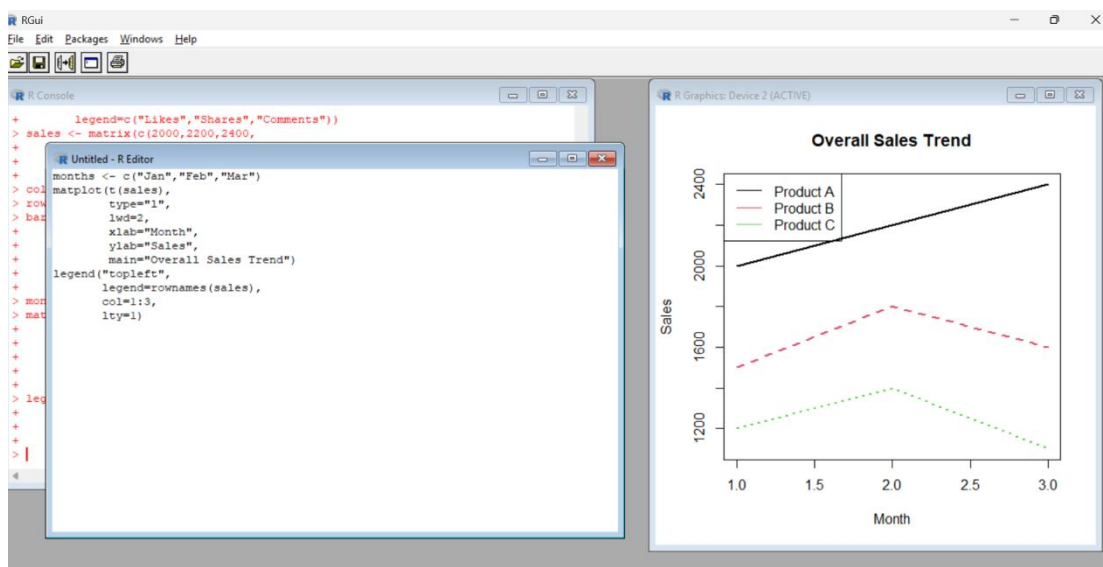


Set 6 : Product Sales Analysis

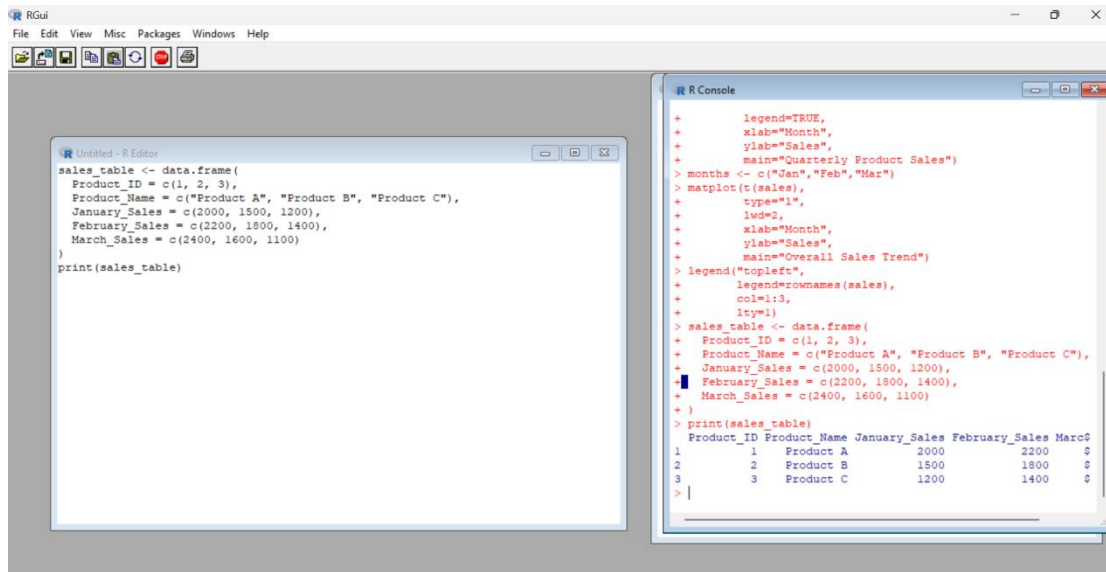
1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.



2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.



3. Build a table to show the monthly sales data for each product. Label the table elements.



4. Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

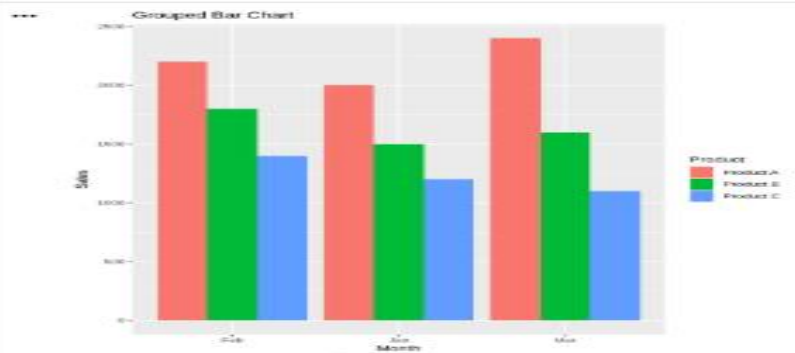
```
library(ggplot2)

data <- data.frame(
  Month = c("Jan", "Jan", "Jan", "Feb", "Feb", "Feb", "Mar", "Mar", "Mar"),
  Product = c("Product A", "Product B", "Product C",
              "Product A", "Product B", "Product C",
              "Product A", "Product B", "Product C"),
  Sales = c(2000, 1500, 1200, 2200, 1800, 1400, 2400, 1600, 1100)
)

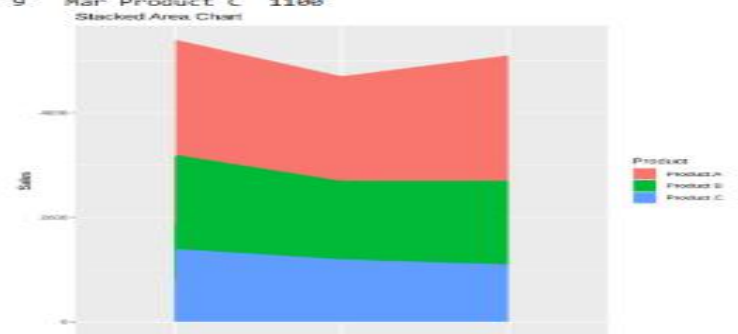
ggplot(data, aes(x = Month, y = Sales, fill = Product)) +
  geom_bar(stat = "identity", position = "dodge") +
  ggtitle("Grouped Bar Chart")

ggplot(data, aes(x = Month, y = Sales, fill = Product, group = Product)) +
  geom_area() +
  ggtitle("Stacked Area Chart")

print(data)
```

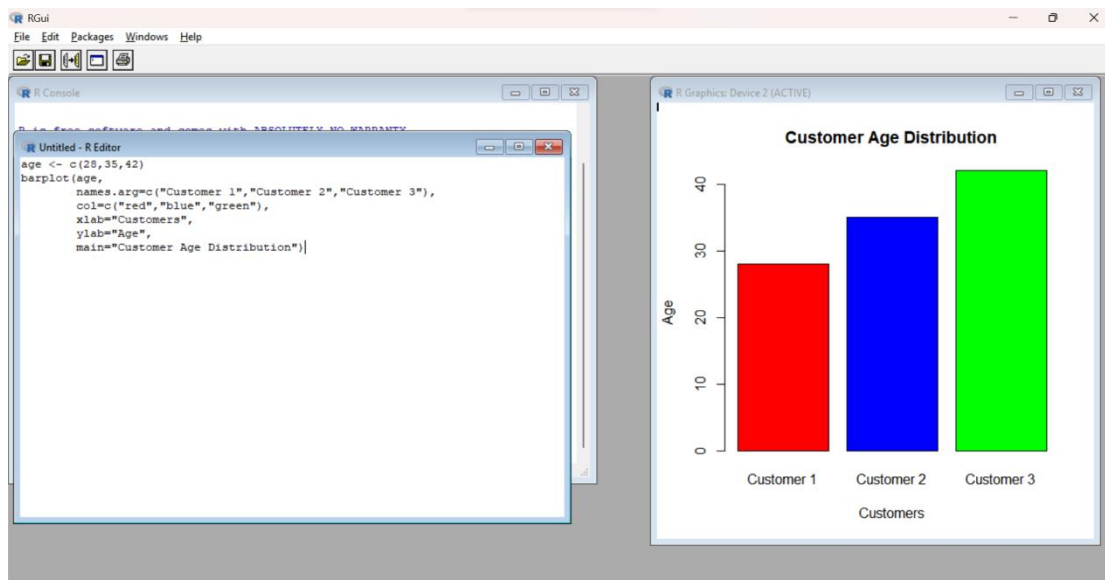


	Month	Product	Sales
1	Jan	Product A	2000
2	Jan	Product B	1500
3	Jan	Product C	1200
4	Feb	Product A	2200
5	Feb	Product B	1800
6	Feb	Product C	1400
7	Mar	Product A	2400
8	Mar	Product B	1600
9	Mar	Product C	1100

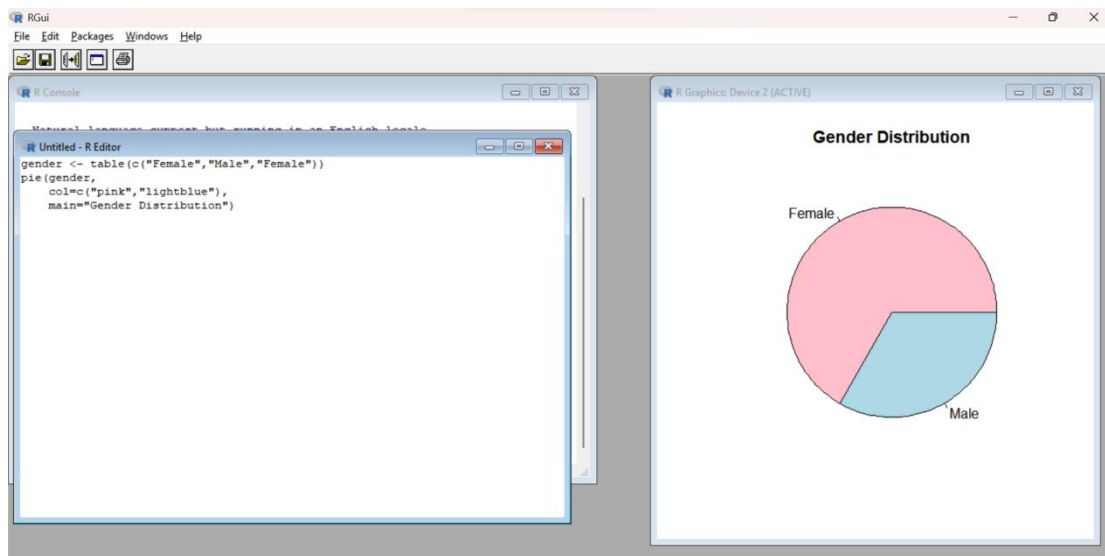


Set 7 Customer Demographics Analysis

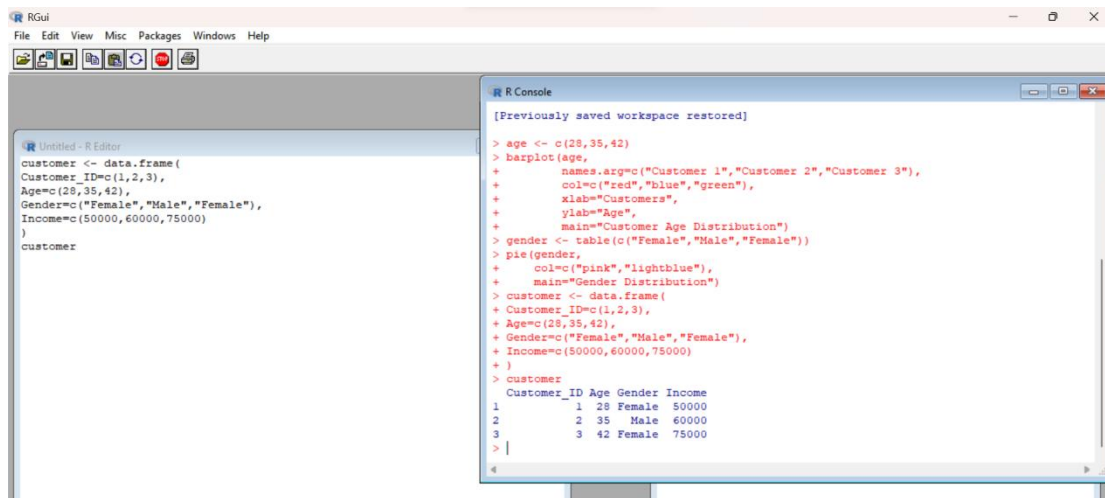
1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.



2. Generate a pie chart to represent the distribution of customers by gender.



3. Build a table to show the demographic information for each customer. Label the table elements.



4. Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

```
library(shiny)
```

```
library(shinydashboard)
```

```
library(ggplot2)
```

```
library(DT)
```

```
customer <-
```

```
  data.frame( CustomerID = c(1, 2,
```

```
  3),
```

```
  Age = c(28, 35, 42),
```

```
  Gender = c("Female", "Male", "Female"),
```

```

Income = c(50000, 60000, 75000)
)
ui <- dashboardPage(
  dashboardHeader(title = "Customer Demographics Dashboard"),
  dashboardSidebar(
    sidebarMenu(
      menuItem("Dashboard", tabName = "dashboard", icon = icon("users"))
    )
  ),
  dashboardBody(

    fluidRow( box
      (
        title = "Customer Age Distribution",
        status = "primary",
        solidHeader = TRUE,
        width = 6,
        plotOutput("barPlot", height = 350)
      ),
      box(
        title = "Gender Distribution",
        status = "success",
        solidHeader = TRUE,
        width = 6,
        plotOutput("piePlot", height = 350)
      )
    ),
    fluidRow( bo
      x(
        title = "Customer Demographics Table",
        status = "warning",

```

```

      solidHeader = TRUE,
      width = 12,
      DTOutput("customerTable")
    )
  )
)

server <- function(input, output)
{ output$barPlot <-
renderPlot({ ggplot(customer,
  aes(x = factor(CustomerID),
    y = Age,
    fill = factor(CustomerID))) +
  geom_bar(stat = "identity") +
  labs(title = "Customer Age Distribution",
    x = "Customer ID",
    y = "Age") + theme_minimal()
  + theme(legend.position =
    "none"))

})

output$piePlot <- renderPlot({
  gender_data <- data.frame(table(customer$Gender))
  colnames(gender_data) <- c("Gender", "Count")
  ggplot(gender_data,
    aes(x = "",
      y = Count,
      fill = Gender)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y") +

```

```

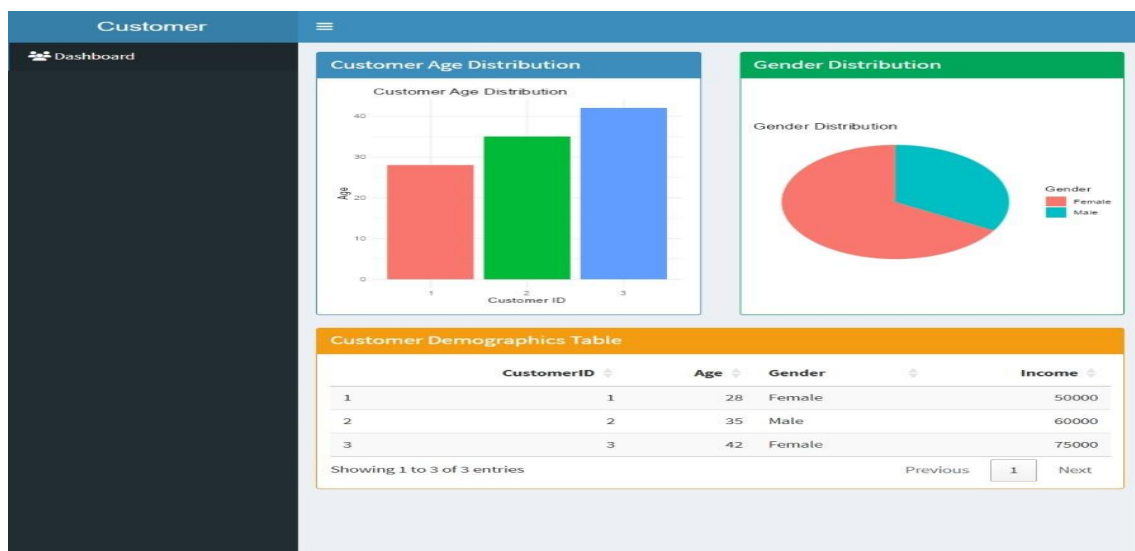
labs(title = "Gender Distribution") +
theme_void()

})

output$customerTable <-
renderDT({ datatable(
  customer,
  options =
list( pageLength
= 5,
  searching = FALSE,
  lengthChange = FALSE
)
)
})
}

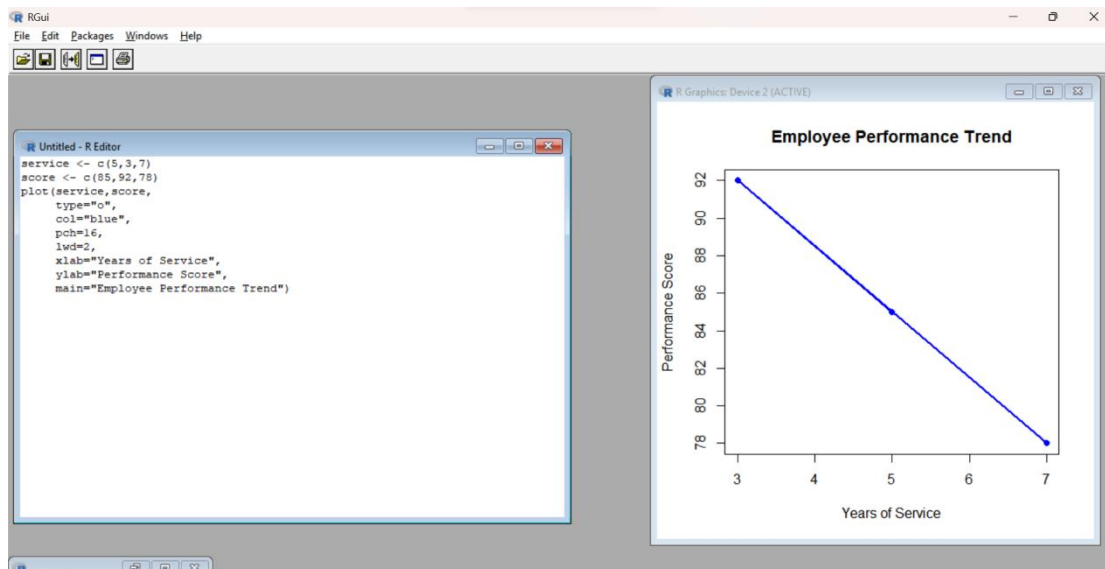
shinyApp(ui, server)

```

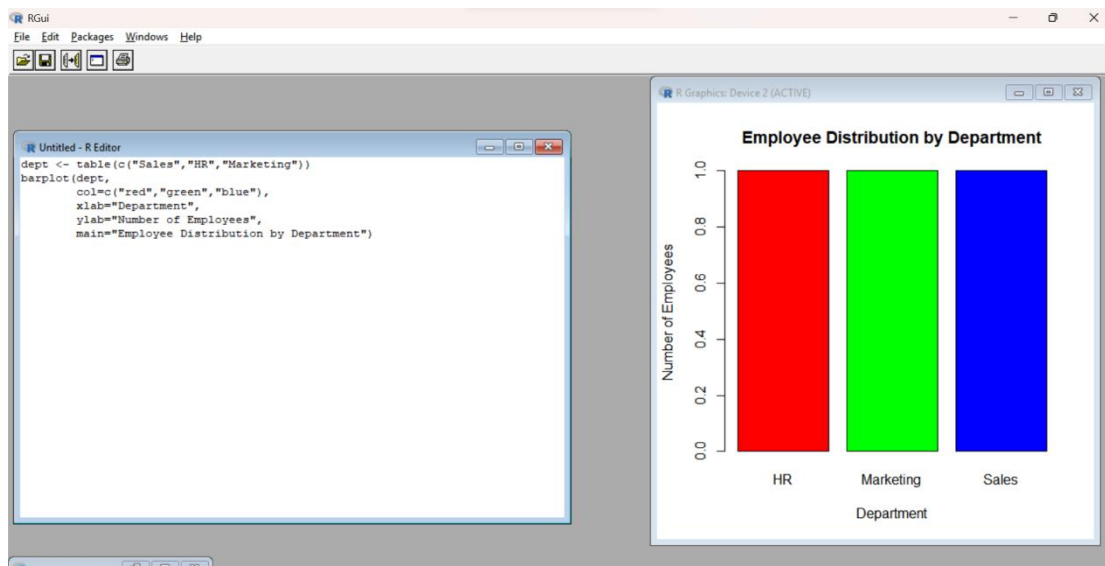


Set 8 Employee Performance Analysis

1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.



2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.



3. Build a table to display the performance data for each employee. Label the table elements.

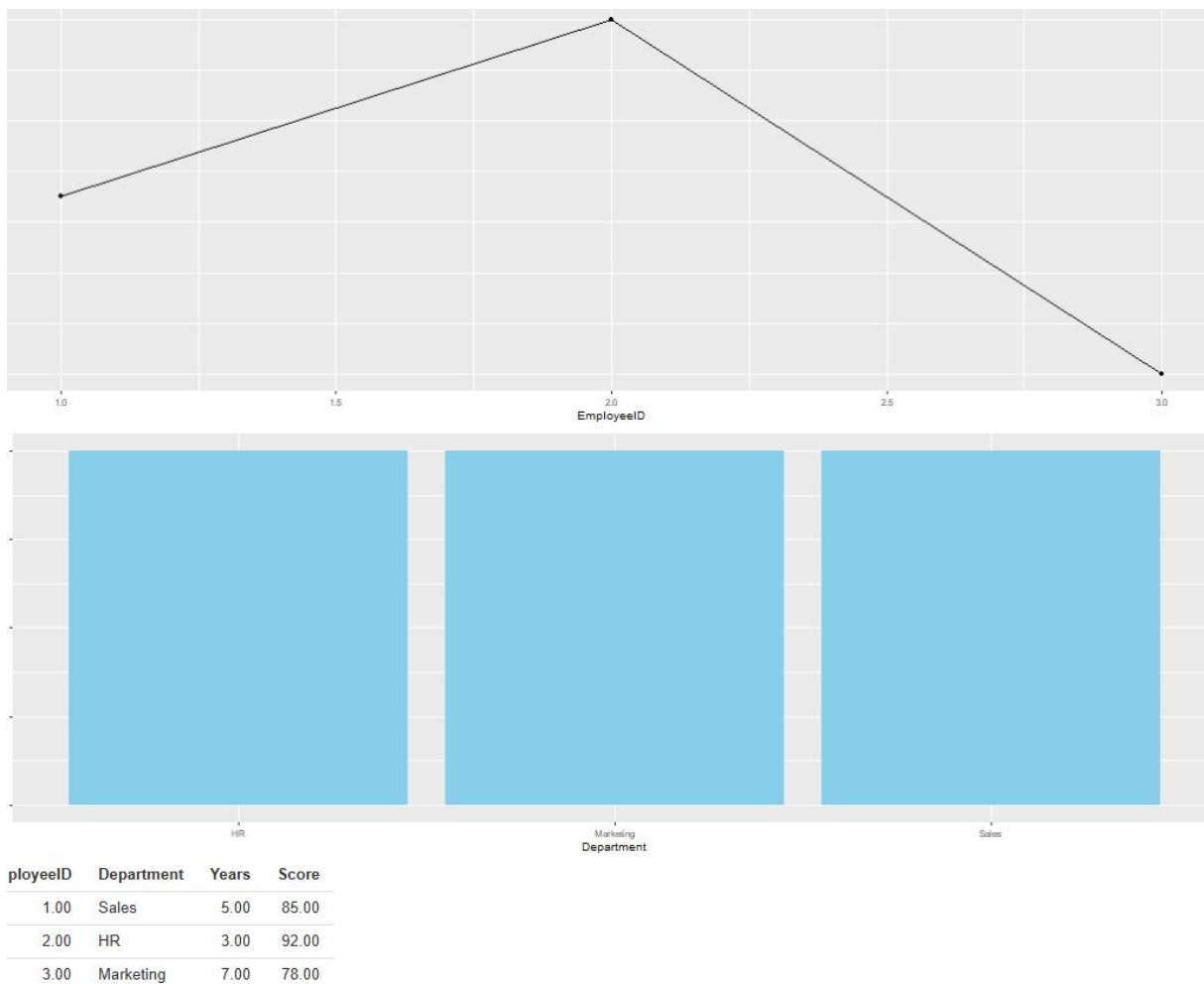
The screenshot shows the RGui interface. The script editor on the left contains the following R code:

```
employee <- data.frame(
  Employee_ID=c(1,2,3),
  Department=c("Sales","HR","Marketing"),
  Years_of_Service=c(5,3,7),
  Performance_Score=c(85,92,78)
)
employee
```

The R Console on the right shows the execution of the code and the resulting data frame:

```
> employee
  Employee_ID Department Years_of_Service Performance_Score
1           1     Sales                5                85
2           2        HR                 3                92
3           3   Marketing                7                78
```

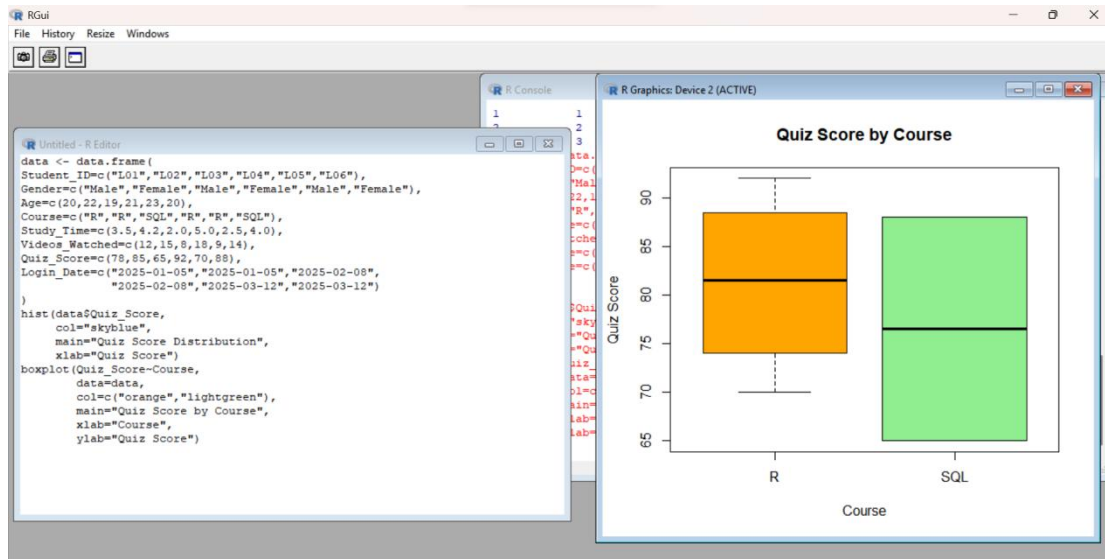
4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.



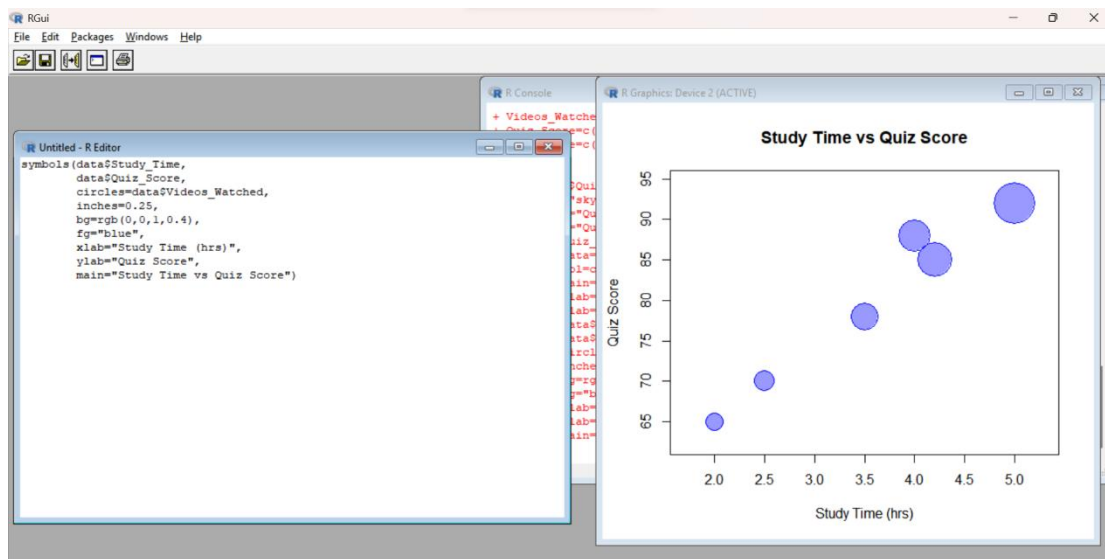
Set 9

Dataset: Online_Learning_Activity.csv

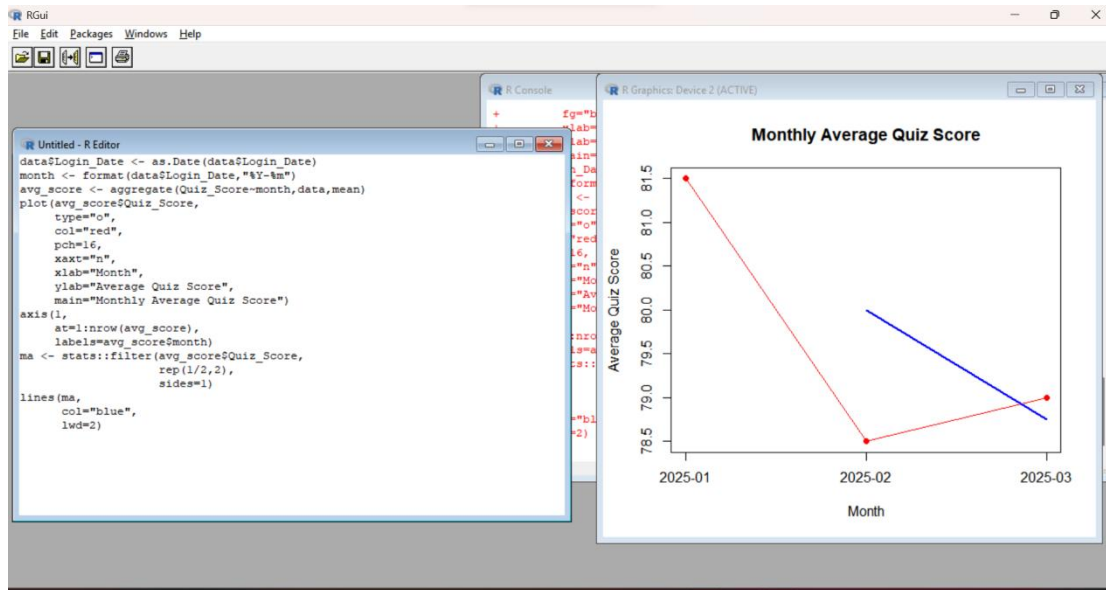
1. Import the dataset in R. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by Course. Add title and labels. Interpret student performance.



2. Create a scatter plot using R programming between Study_Time and Quiz_Score. Represent Videos_Watched using bubble size. Apply transparency. Explain the learning pattern.



3. Using R programming, convert Login_Date into Date format. Calculate average quiz score per month. Plot a line chart. Apply moving average smoothing. Identify the trend.



4. Import Online_Learning_Activity.csv into Tableau. Create a bar chart showing average quiz score by Course. Create a scatter plot (Study Time vs Quiz Score). Add filter for Gender and Course. Design a mini interactive dashboard.

```
library(shiny)
library(ggplot2)

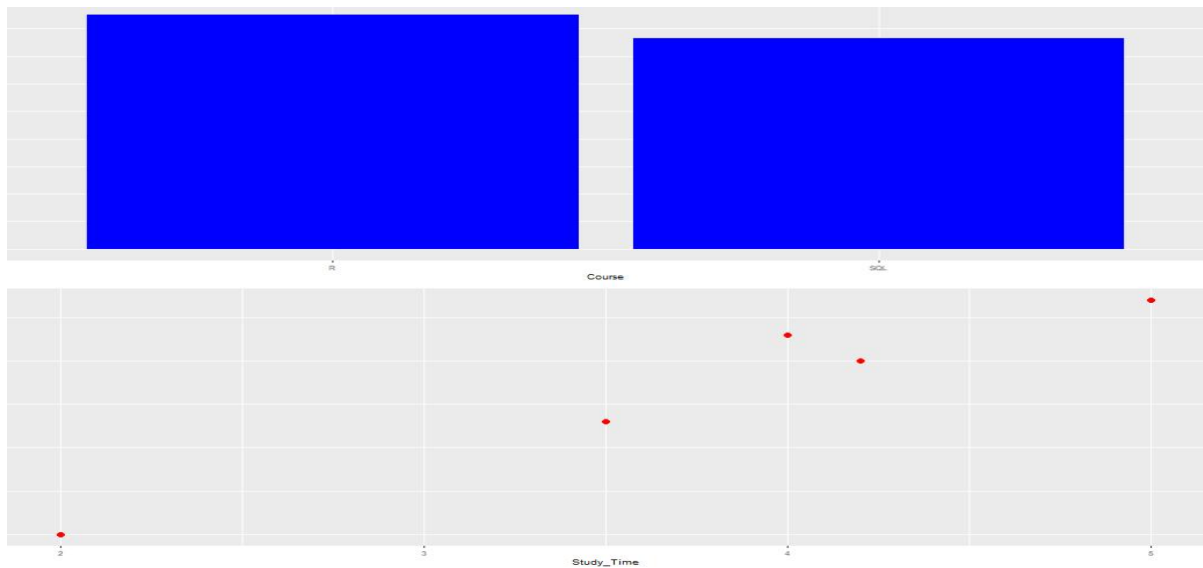
data <- data.frame(
  Course = c("R","R","SQL","R","SQL"),
  Quiz_Score = c(78,85,65,92,88),
  Study_Time = c(3.5,4.2,2.0,5.0,4.0)
)

ui <-
  fluidPage( plotOutput("
  bar"),
  plotOutput("scatter"
  )
server <- function(input,
  output){ output$bar <-
  renderPlot({
    ggplot(data, aes(Course, Quiz_Score)) +
      stat_summary(fun=mean, geom="bar", fill="blue")
  })
```

```

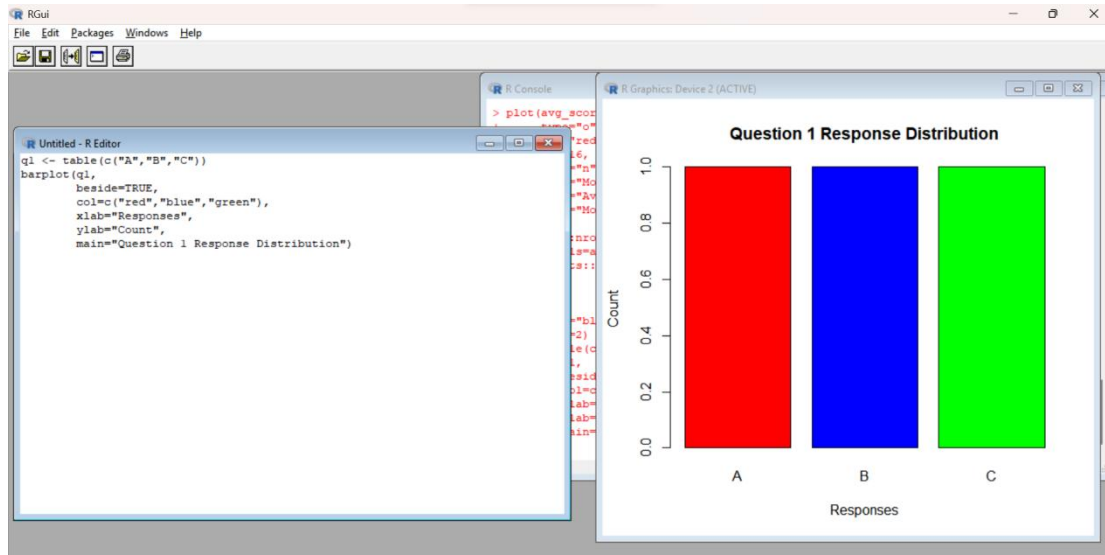
output$scatter <- renderPlot({
  ggplot(data, aes(Study_Time, Quiz_Score)) +
    geom_point(color="red", size=3)
})
}
shinyApp(ui, server)

```

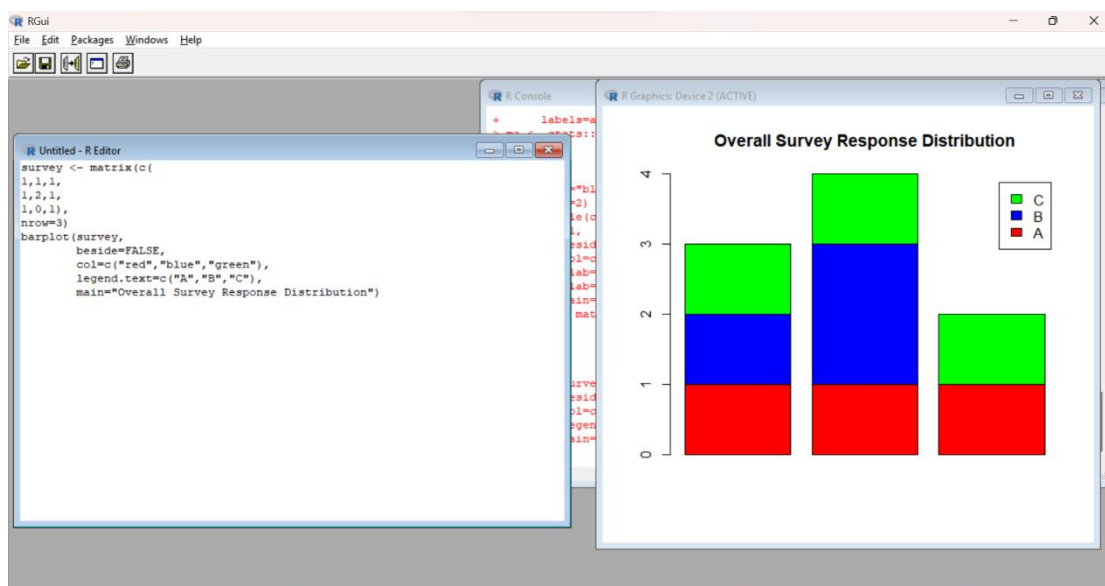


Set 10 Survey Responses Analysis

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.



2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.



3. Build a table to show the survey response data for each survey. Label the table elements.

```
library(shiny)
library(ggplot2)

main="Question 1 Response Distribution"
> survey <- matrix(c(
+ 1,1,1,
+ 1,2,1,
+ 1,0,1),
+ nrow=3)
> barplot(survey,
+         beside=FALSE,
+         col=c("red","blue","green"),
+         legend.text=c("A","B","C"),
+         main="Overall Survey Response Distribution")
> survey_data <- data.frame(
+ Survey_ID=c(1,2,3),
+ Question1=c("A","B","C"),
+ Question2=c("B","A","A"),
+ Question3=c("C","D","B")
+ )
> survey_data
  Survey_ID Question1 Question2 Question3
1         1         A         B         C
2         2         B         A         D
3         3         C         A         B
```

4. Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.

```
library(shiny)
```

```
library(ggplot2)
```

```
data <- data.frame(
```

```
  SurveyID = c(1,2,3),
```

```
  Q1 = c("A","B","C"),
```

```
  Q2 = c("B","A","A"),
```

```
  Q3 = c("C","D","B")
```

```

)
ui <- fluidPage(
  h2("Survey Responses Dashboard"),
  plotOutput("bar1"),
  plotOutput("bar2"),
  tableOutput("table")
)
server <- function(input,
  output){ output$bar1 <-
  renderPlot({ ggplot(data, aes(Q1))
  +
    geom_bar(fill="skyblue") +
    ggtitle("Grouped Bar Chart")
  })
  output$bar2 <-
  renderPlot({ df <-
  data.frame(
    Question = rep(c("Q1", "Q2", "Q3"), each=3),
    Response = c(data$Q1, data$Q2, data$Q3)
  )
  ggplot(df, aes(Question, fill=Response)) +
    geom_bar() +
    ggtitle("Stacked Bar Chart")
  })
  output$table <-
  renderTable({ data
  })
})
shinyApp(ui, server)

```

Survey Responses Dashboard

